MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE 6BM6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, separate cavity, pulse, and CW operation

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ik	Ehk	Pi	tk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	Vdc	W	sec	ft
Maximum:	6.9	+1.0	350	350	-700	32	+45	10	---	10,000
Minimum:	5.7	-500	---	---	-15	---	-45	---	120	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Coated unipotential
Mounting position:	Any

TEST CONDITIONS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er
Unit:	V	Vdc	Vdc	Vdc	Vdc
Test condition 1:	6.3	0	325	325	Adj for max Po
Test condition 2:	6.3	0	300	300	Adj for max Po

GENERAL:

Qualification - Required

1/ CAUTION: Tube not designed for applications where hysteresis may affect performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:35Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	-					
4229	Total reflector current (1)	4	-	Er = -90 Vdc	Ir	---	5	µAdc
1301	Heater current	-	-		If	500	750	mA
1336	Heater-cathode leakage	-	-	Ehk = ±45 Vdc	Ihk	0	75	µAdc
1256	Electrode current (1) (cathode)	5	1		Ik	13	29	mAdc
4213	Reflector voltage (1)	5	1		Er	-30	-70	Vdc
4229	Total reflector current (2)	5	1		Ir	---	150	µAdc
---	Frequency (1)	5	1		F	785	815	MHz
4250	Power output (1)	5	1		Po	50	---	mW
4214	Cathode emission (1)	5, 6	1		ΔIk/Ik	---	15	%
1256	Electrode current (2) (cathode)	7	1		Ik	13	29	mAdc
4213	Reflector voltage (2)	7	1		Er	-450	-550	Vdc
---	Frequency (2)	7	1		F	2,150	2,250	MHz
4250	Power output (2)	7	1		Po	50	---	mW
4214	Cathode emission (2)	6, 7	1		ΔIk/Ik	---	15	%
1256	Electrode current (3) (cathode)	8	2		Ik	13	29	mAdc
4213	Reflector voltage (3)	8	2		Er	-20	-60	Vdc
4229	Total reflector current (3)	8	2		Ir	---	150	µAdc
---	Frequency (3)	8	2		F	1,130	1,170	MHz
4250	Power output (3)	8	2		Po	10	---	mW
4214	Cathode emission (3)	6, 8	2		ΔIk/Ik	---	15	%
1256	Electrode current (4) (cathode)	9	2		Ik	13	29	mAdc
4213	Reflector voltage (4)	9	2		Er	-275	-355	Vdc
---	Frequency (4)	9	2		F	2,680	2,820	MHz
4250	Power output (4)	9	2		Po	20	---	mW
4214	Cathode emission (4)	6, 9	2		ΔIk/Ik	---	15	%

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:35Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2							
1101	Secureness of base, cap, or insert	3	-		---	---	---	---
1031	Low-frequency vibration	-	-	No voltages	---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	2	1	Group C	t	500	---	hrs
---	Life-test end points:							
4250	Power output (3)	8	2		Po	8	---	mW
4250	Power output (4)	9	2		Po	16	---	mW

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. Test tube with 1,000 MHz test cavity as specified on Drawing 166-JAN. Frequency approximately 1,000 MHz. Tube operated in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
3. Water-immersion test omitted. Torque shall be applied between upper disk (g3) and molded portion of base and shall be 20 inch-pounds.
4. Static test; tube not oscillating.
5. Test tube with 800 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 800.5 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
6. Reduce Ef to 5.7 V while tube is oscillating. The cathode emission shall not exceed the limit shown.
7. Test tube with 2,200 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 2,113.8 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 1 mode (1-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
8. Test tube with 1,150 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 1,130.6 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
9. Test tube with 2,750 MHz test cavity as specified on Drawing 166-JAN. Adjust cavity to resonate at 2,569.2 MHz with standard plug as specified on Drawing 197-JAN. Tube operated in N = 2 mode (2-3/4 cycles for complete transit). Adjust Er for maximum power output. Adjust output coupling loop for maximum power output.
10. Changes from previous issue. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:35Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

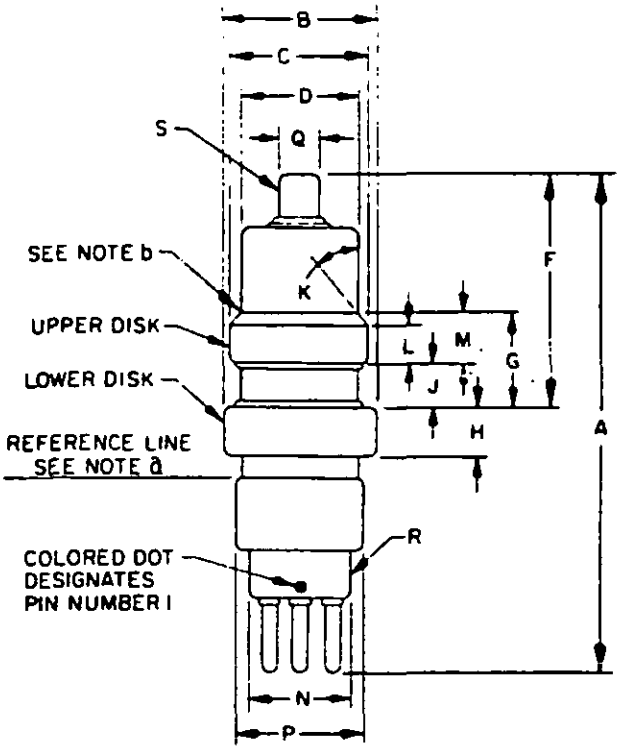
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2745)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:35Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	2.969 (75.41)	3.219 (81.76)
B	.990 (25.15)	1.000 (25.40)
C	.880 (22.35)	.890 (22.61)
F	1.375 (34.93)	1.500 (38.10)
M	.303 (7.70)	.323 (8.20)
Quality conformance inspection, part 2		
G	.560 (14.22)	.610 (15.49)
H	.250 (6.35)	.312 (7.92)
J	.257 (6.53)	.287 (7.29)
Quality conformance inspection, part 3		
D		.750 (19.05)
K	25°	35°
N	.610 (15.49)	.656 (16.66)
P	.807 (20.50)	.822 (20.88)
Q	.245 (6.22)	.255 (6.48)
R	Base: A4-76	
S	Cap: C1-3	
Nominal dimensions		
L	.250 (6.35)	



PIN CONNECTIONS	
1	g1
2	h
3	k
4	h
Lower disk	g2
Upper disk	g3
Cap	r

NOTES:
a. For dimensions below reference line refer to base A4-76.
b. Space between glass and upper disk edge or backing ring shall be .030 maximum. Quality conformance inspection, part 2 shall apply.
c. Diameters B and C shall be concentric within .025 TIR and shall form circular cylinder within a range of .006. Allowable radius at ends of cylinder shall be .030 maximum. Quality conformance inspection, part 2, shall apply.
d. Tube shall be tested in concentricity gage specified on drawing 199-JAN. Gage readings on diameters shall be as follows:

Diameter	Eccentricity
D	.050 max
N	.060 max
P	.060 max
Q	.040 max

e. All concentricity measurements shall be quality conformance inspection, part 2.
f. Pin No. 1 shall be identified by an index mark (colored dot or other).

FIGURE 1. Outline drawing of electron tube type 6BM6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:35Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:35Z
Check the source to verify that this is the current version before use.